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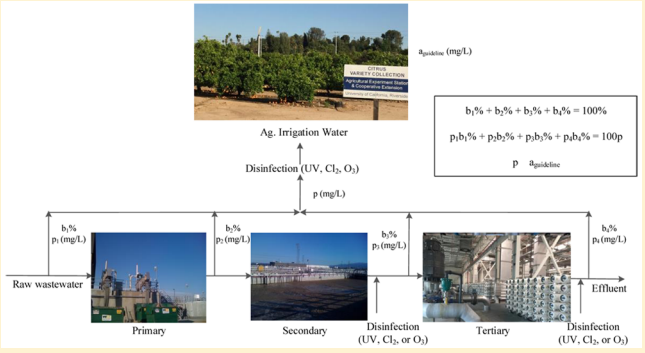
Wastewater Reuse for Agriculture: Development of a Regional Water Reuse Decision-Support Model (RWRM) for Cost-Effective Irrigation Sources

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Supporting Information

ABSTRACT: Water scarcity has become a critical problem in many semiarid and arid regions. The single largest water use in such regions is for crop irrigation, which typically relies on groundwater and surface water sources. With increasing stress on these traditional water sources, it is important to consider alternative irrigation sources for areas with limited freshwater resources. One potential irrigation water resource is treated wastewater for agricultural fields located near urban centers. In addition, treated wastewater can contribute an appreciable amount of necessary nutrients for plants. The suitability of reclaimed water for specific applications depends on water quality and usage requirements. The main factors that determine the suitability of recycled water for agricultural irrigation are salinity, heavy metals, and pathogens, which cause adverse effects on human, plants, and soils. In this paper, we develop a regional water reuse decision-support model (RWRM) using the general algebraic modeling system to analyze the cost-effectiveness of alternative treatment trains to generate irrigation water from reclaimed wastewater, with the irrigation water designed to meet crop requirements as well as California's wastewater reuse regulations (Title 22). Using a cost-minimization framework, least-cost solutions consisting of treatment processes and their intensities (blending ratios) are identified to produce alternative irrigation sources for citrus and turfgrass. Our analysis illustrates the benefits of employing an optimization framework and flexible treatment design to identify cost-effective blending opportunities that may produce high-quality irrigation water for a wide range of end uses.



INTRODUCTION

With concerns over water availability and reliability, municipalities and state governments are focusing more attention and resources on the reuse of municipal wastewater. California's Orange County Water District (OCWD), which operates the largest groundwater replenishment system in the world, has invested $142 million to increase its current wastewater recycling plant capacity from 70 to 100 million gallons per day (MGD).[1] Meanwhile, the neighboring Metropolitan Water District of Southern California (MWD), the country's largest water district, is considering investing $15 million in a demonstration project that is intended to pave the way for the development of a 150 MGD wastewater recycling plant.[2] Such investments are consistent with national trends, as revealed in a recent National Academy of Sciences report, which notes that the volume of wastewater reused in the United States is increasing at an annual rate of 15%.[3] California, Texas, and Arizona, three of the top four states in terms of total volume reused, are located in the arid southwest and are expected to experience more frequent and intense droughts under climate change.[4]

Although the reuse of treated wastewater is not a new concept, concerns over the rising demand for water from population growth, coupled with both economic and environmental challenges, have made this option more attractive.[3,5,6] For instance, the estimated costs for MWD to recycle wastewater is approximately $0.72/m^3 ($900 per acre-ft), significantly less than the current cost of $1.13/m^3 ($1400 per acre-ft) for imported water from northern California via the State Water Project (SWP), which supplies approximately 30% of water used in Southern California.[2] In addition to offering a less-costly alternative to imported water, reuse of wastewater is also seen as a more environmentally friendly and locally dependable option, particularly during drought when the reliability of imported supplies decreases substantially. Indeed, during the drought confronting California in 2014, one of the driest periods in California history, only 5% of the SWP allocation water was delivered to agencies in Southern

Received: April 26, 2016
Revised: August 6, 2016
Accepted: August 7, 2016
Published: August 8, 2016

ACS Publications © 2016 American Chemical Society

DOI: 10.1021/acs.est.6b02073
Environ. Sci. Technol. 2016, 50, 9390−9399

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California, as compared to 80% in 2011.[7] Because farmers in California are heavily reliant on SWP deliveries for irrigation, low allocations mean farmers must turn to other sources. The reuse of treated wastewater for irrigation, which constitutes 80−90% of the nation's consumptive water use (water lost to the environment through evaporation, crop transpiration, or incorporation into products), is one such possible source that presents water agencies and farmers with the possibility of a low-cost, reliable, and environmentally friendly local water source whose value will only increase under expected climate change conditions.[8]

Currently, most wastewater treatment plants in California treat municipal wastewater to secondary or tertiary standards, followed by disinfection and discharge of the treated effluent to a surface water body.[6] Increasingly, disinfected secondary and tertiary effluents are used to irrigate restricted and unrestricted access areas, such as golf courses and freeway medians. The degree to which treated wastewater may broaden the water supply portfolios of particular water agencies and substitute for other water supply sources depends largely upon the chemical and biological composition of the effluent. Different water quality parameters (which often vary geospatially) can impact the suitability of recycled wastewater for irrigation, including salinity (expressed as total dissolved solids (TDS)), nutrient load, heavy metals, and pathogens, which might cause adverse effects on human, plants, and soil health.[5,9−12] The concentrations of these elements in wastewater impacts the treatment approach, and hence cost, necessary to generate irrigation water of appropriate quality.

Yet such limitations on the expanded use of treated wastewater need not be the case. For example, the removal of salt would generate usable irrigation water that meets minimum salinity thresholds for crops.[9] For this alternative treatment, then, some degree of desalination is necessary.[13] Although adding desalination will increase costs, recycled wastewater can contribute an appreciable amount of the necessary nutrients for plants, which reduces the need for (and costs associated with) synthetic fertilizers.[14] Thus, as the demand for treated wastewater increases from different sectors of society (e.g., agricultural, municipal potable and nonpotable, and environmental sectors), the benefits of tailoring wastewater treatment plant operations to the intended use can reduce unnecessary expenditures on capital and operations and maintenance (O and M) expenditures, energy, and greenhouse gas emissions and provide utilities with a more-competitive option relative to other water supply sources.[15] As such, a flexible treatment approach that pairs specific wastewater treatment steps (arranged in treatment trains) and resulting water quality, with specific crop water quality demands, might offer utilities at the urban−agricultural interface a cost-effective means of transforming wastewater from a waste product to a valuable commodity.

The objective of this paper is to illustrate, through the development of a regional water reuse decision-support model (RWRM), how flexible wastewater treatment processes that allow blending can be optimized to produce a water supply that meets and surpasses a variety of water quality requirements at an affordable cost. While regulations such as Title 22 tend to focus on pathogen removal to ensure safe reuse of wastewater, this study instead focuses on producing irrigation water with tailored *chemical* properties, with the assumption that a disinfection step is needed to maintain safe wastewater reuse. In particular, we estimate and compare the costs and water quality characteristics of treated wastewater under a wide array of feasible treatment combinations that meet irrigation guidelines for two different types of products: citrus and turfgrass. The model, which has been developed and calibrated to reflect current wastewater treatment plant processes, costs, irrigation guidelines, and regulatory requirements in California, also identifies the cost-effective solution under alternative nutrient and bicarbonate constraints. Our solutions illustrate that when wastewater treatment management matches the water quality requirements of certain crops with an optimized wastewater treatment train that ensures the crops receive irrigation water tailored to their specific needs, an affordable water supply can be created that can meet and even exceed current water quality standards and practices. Consequently, the key contribution of this paper is to identify cost-effective blending combinations across treatment processes that maintain crop yield and long-term soil health.

MATERIALS AND METHODS

Modeling Methodology. Here, we discuss the modeling procedures and input data used to analyze the cost-effectiveness of alternative treatment trains to generate irrigation water from wastewater while meeting specific crop requirements. A unique element of our model is its ability to consider and identify cost-effective blending combinations across processes and the resultant water quality parameters that meet crop requirements. That is, rather than irrigating crops with one type of recycled water, e.g., disinfected secondary or tertiary effluents, the model provides water managers the opportunity to identify blended streams with higher quality at a lower price.[5] The blended effluents can be used on different crops, depending on specific irrigation guidelines.

Current wastewater treatment practices often consist of exposing the entire volume of influent to secondary or tertiary treatment prior to discharge.[16] Our analysis consists of considering the cost-effectiveness of a more-flexible system in which fractions of plant influent, primary, secondary (membrane bioreactors (MBR) or activated sludge), and tertiary (filtration, membrane separation, and desalination) effluents are combined to produce a blend that meets particular demand criteria. When the demand for the effluent is for irrigation of particular crops (e.g., citrus or turfgrass), as it is in our current analysis, the criteria are based on plant requirements (i.e., nutrients) and thresholds (i.e., bicarbonate and salinity) along with any additional regulatory restrictions (such as a disinfection requirement) at the least cost. For example, under California's Title 22 regulations, recycled water used for the irrigation of food crops (where the recycled water comes in contact with the edible portions of the crops) and unrestricted access golf courses must meet tertiary-treated quality in which the total coliform concentration must exceed neither a 7 day median measure of 2.2 MPN/100 mL nor a 23 MPN/100 mL value in more than one sample in any 30 day period.[17]

The RWRM was evaluated by analyzing three different scenarios corresponding to different model constraints: (a) with nutrient (N, P, K) and bicarbonate constraints (baseline), (b) without crop nutrient constraints, and (c) without nutrient and bicarbonate constraints. Crop nutrients and bicarbonate constraints were developed based on published data regarding the average amount of nutrients and bicarbonate in irrigation water (i.e., irrigation guidelines) typically used for each crop, which, through long-term experimental projects, caused no

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Environ. Sci. Technol. 2016, 50, 9390−9399

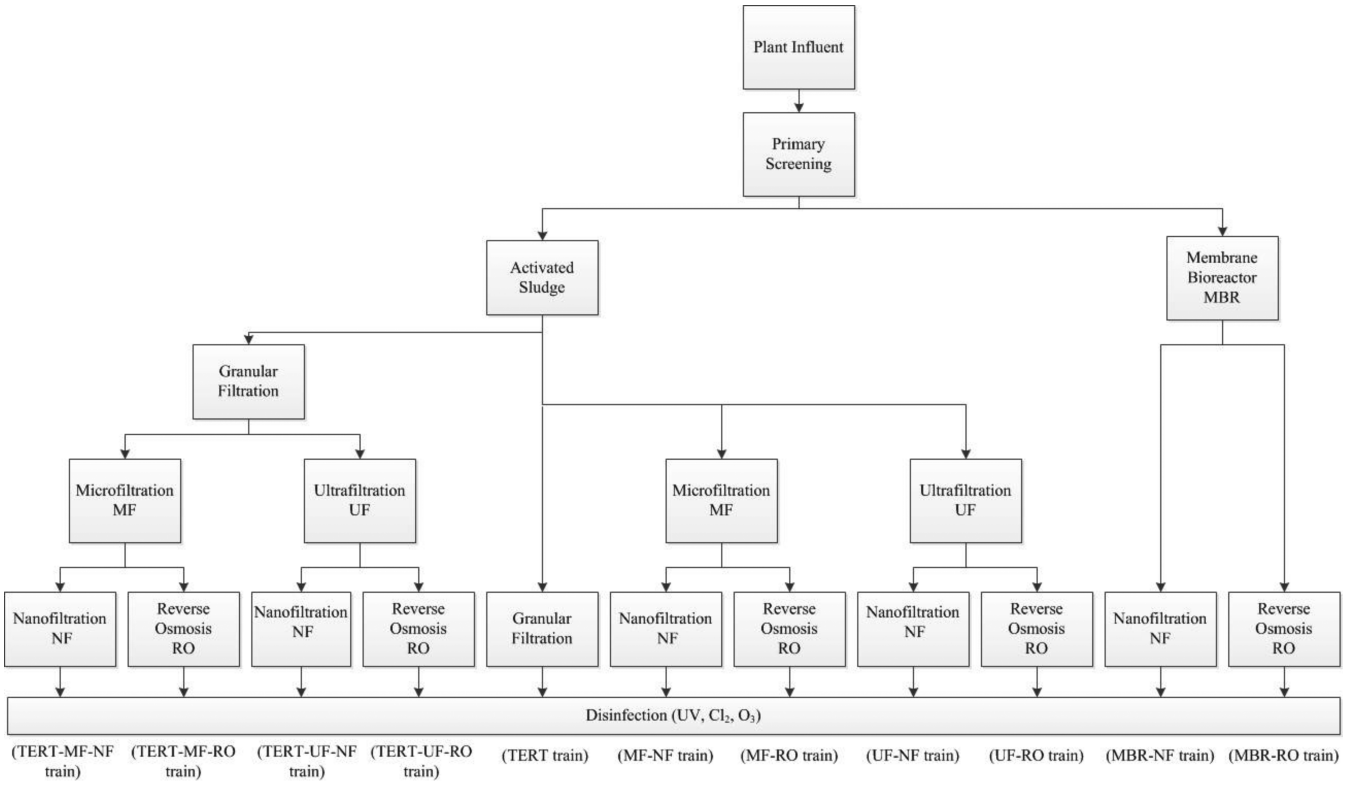


Figure 1. Different treatment train processes corresponding to different GAMS logic.

adverse effects on crop yield and soil health.[9,18] Highly concentrated bicarbonate in irrigation water can cause soil permeability problems.[9] However, the bicarbonate concentration in typical municipal wastewater is, in fact, below the maximum allowable bicarbonate concentrations suitable for citrus and turfgrass (less than 8.5 mequiv/L or 519 mg/L HCO_3^- for moderate restriction).[9,18] While 519 mg/L of bicarbonate concentration is far higher than what is typically found in wastewater (and irrigation water), this level of bicarbonate will result in a decrease in crop yield, which is beyond the scope of our current modeling effort.[12] We include the nutrients and bicarbonate constraints here (which are, in essence, a conservative design meant to represent concentration levels of these factors typical in effluent) to evaluate the sensitivity of the model to different blending combinations as well as the treatment costs associated with varying restrictions on input parameters. With the increase of nutrient concentrations in terms of NH_4–N, NO_3–N, PO_4–P, and K in irrigation water, i.e., the removal of the nutrient constraints, crop demand for synthetic fertilizers can be reduced, which offers irrigators further possibilities to reduce costs.

We begin by discussing the alternative treatment trains that comprise our analysis, followed by a presentation of the optimization framework employed to identify the cost-effectiveness of each possible train. Results showing optimal blending ratios and cost savings associated with the reduced use of synthetic fertilizers for specific crops (citrus and turfgrass) are then highlighted. Data on the effectiveness of each process and its costs are reported as well.

Development of Regional Water Reuse Decision-Support Model (RWRM) and Optimization Framework. The RWRM was developed by focusing on combinations of treatment processes that are compatible with one another while avoiding redundancy. A total of seven feasible wastewater treatment technologies, arranged in 11 different wastewater treatment trains, were identified (Figure 1). These trains include: (1) activated sludge–granular filtration (TERT); (2) activated sludge–microfiltration–RO (MF-RO); (3) activated sludge–MF–nanofiltration (MF–NF); (4) activated sludge–ultrafiltration–RO (UF–RO); (5) activated sludge–UF–NF (UF–NF); (6) MBR–RO; (7) MBR–NF; (8) activated sludge–granular filtration–MF–RO (TERT–MF–RO); (9) activated sludge–granular filtration–MF–NF (TERT–MF–NF); (10) activated sludge–granular filtration–UF–RO (TERT–UF–RO); and (11) activated sludge–granular filtration–UF–NF (TERT–UF–NF). While typical wastewater treatment plants do not employ a desalination step (NF or RO), this technology is emerging in certain instances in which direct or indirect potable reuse is implemented. However, in contrast to potable reuse, to produce irrigation quality water, only a portion of the effluent stream may require desalination, which will depend on the water quality characteristics of the particular wastewater stream. When the blending ratios (i.e., the proportion of water in a given volume from each treatment process) are evaluated, the final irrigation water can be composed of effluent from any of the technologies in a treatment train. For example, the final blended water could be composed of primary effluent, secondary effluent, and RO effluent (or any other combination). These 11 treatment trains represent current municipal wastewater treatment technologies and cover almost all possible treatment configurations. Each of these trains provides for the removal of nutrients (carbon, nitrogen, and phosphorus) through the activated sludge or MBR process and the removal of large particles and molecules along with bacteria and viruses through the use of the MF and UF processes. The NF and RO processes, meanwhile, effectively remove divalent and monovalent ions, respectively.

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Environ. Sci. Technol. 2016, 50, 9390–9399

The removal efficiencies of different wastewater constituents by the different treatment steps are provided in Table S1.

The cost of each treatment train was estimated as the product of the unit treatment cost ($/m^3) for individual processes within a particular train multiplied by the blending fraction of the produced water associated with that process, as shown in eq 1:

$$C_T = \sum_{j=1}^{J_T} b_T(j)^* c_T(j) \quad (1)$$

where C_T is the total treatment cost per m^3 of wastewater treated, T is the particular treatment train, and j is the specific process within a treatment train T, which consists of J_T unique processes. The blending ratio of the produced water associated with process j for treatment train T is designated by $b_T(j)$, whereas the unit cost of each process is $c_T(j)$. The treatment cost is subject to two constraints. First, the total blending fractions of the produced water from each treatment train is restricted to one, corresponding to 100% of a given effluent volume as in eq 2, assuming all effluent is used by the crop.

$$\sum_{j=1}^{J_T} b_T(j) = 1 \quad (2)$$

Second, crop-specific upper limits are imposed on irrigation water quality parameters from each treatment train. The final concentrations of constituents resulting from the blending process, as shown in eq 3, must be less than those specified in a certain crop's irrigation guidelines, assuming water quality parameter values are additive:

$$\sum_{j=1}^{j_T} p_T(\mathrm{i}, j)^* b_T(j) \le a(\mathrm{i}) \quad (3)$$

where i is a specific wastewater constituent associated with process j within a given treatment train T, $p_T(i,j)$ is the concentration of specific constituent i resulting from process j, and a(i) is the concentration threshold for constituent i so as to not impact crop yield.

Using the General Algebraic Modeling System (GAMS), the lowest-cost solution is identified by choosing the blending ratios and treatment train that minimize the cost of producing a particular volume of effluent under the blending and irrigation guideline constraints. Our optimization framework, thus, seeks to minimize $C_T = \Sigma_{j=1}^{J_T} b_T(j)^* c_T(j)$ subject to eqs 2 and 3.

Model Inputs. Model inputs are composed of four categories: concentrations of water quality parameters in a given unit volume of wastewater, concentrations of water quality parameters associated with effluent from each treatment process, concentration limits of water quality parameters to meet crop requirements, and the unit cost for each treatment process. Wastewater influent, or raw wastewater, typically contains high levels of nutrients, specifically nitrogen (NH_4–N, 40.3 mg/L), phosphorus (P, 9.7 mg/L), and potassium (K, 15.9 mg/L), which can be utilized by plants.[19,20] However, to meet state and federal water quality regulations, most conventional municipal wastewater treatment plants subject this raw influent to primary, secondary, tertiary, and disinfection processes, which results in significant removal of nutrients (typical treated effluent concentrations of 5.2 mg/L NH_4–N and 6.6 mg/L P).[5,20−22] However, wastewater may contain a high load of TDS, which are not effectively removed by conventional treatment processes.[22] To reduce salinity and meet more-stringent effluent quality regulations, membrane-based treatment methods are becoming more common in wastewater treatment.[22] Membrane separation processes commonly used in water treatment processes include MF, UF, NF, and RO. These processes rely on physical (size exclusion and charge exclusion) and chemical (differences in diffusion rates) phenomena to remove contaminants from waste streams.[23]

A summary of water quality parameters in the wastewater, as well as the concentration after each treatment process, can be found in Table S2. These water quality parameters are represented by $p_T(i,j)$ in eq 3. Average removal rates of contaminants by each treatment process are displayed in Table S1. The data in Table S2 were obtained from sources including the Inland Empire Utilities Agency, the U.S. EPA, and OCWD.[19,20,22,24] In instances where data were not available (italicized entries in Table S2), parameters were estimated based on typical removal percentages (e.g., MF and UF are known to not effectively remove TDS). RO results in the lowest concentrations of salts, nutrients, and other constituents compared to NF, UF, and MF. It is important to note that TDS levels in the model wastewater used here (522 mg/L in secondary effluent and 496 mg/L in tertiary effluent) are near or below the maximum recommended TDS values for citrus and most turfgrass. Thus, salinity was not a limiting factor when determining the optimal blending ratios. However, many wastewater plants generate high-salinity (TDS > 500 mg/L) effluents. For example, the wastewater from the city of Carlsbad in southern California contains TDS levels in excess of 1000 mg/L.[25]

Specific water quality standards and thresholds associated with irrigation water quality parameters for citrus and turfgrass can be found in Table S3;[9,22,26−28] the concentration thresholds for each parameter correspond to the a(i) values in eq 3. In general, citrus has lower threshold concentrations than turfgrass. For comparison purposes, Table S3 also includes the range of concentrations typically found in standard irrigation water. In considering the model outcomes (left side of eq 3), many blending ratios (model solutions) generated water with high levels of pathogens. In such instances, an additional disinfection step using chlorine, ozone, UV radiation, or some or all of these was necessary to meet California Title 22 regulations for water reuse on agriculture.

A summary of estimated individual treatment costs associated with the individual processes within a treatment train for small−medium (≤20 000 m^3/day (5 MGD)) and large (>20 000 m^3/day) treatment facilities is provided in Table S4. Capital costs were amortized over 15 to 20 years, depending on the process, and included construction and administrative costs. Operation and maintenance costs reflect the cost of energy, chemicals, and maintenance.[29] The information in Table S4 spans several plant capacities and time intervals, with data adjusted to 2013 U.S. dollars using the Engineering News Record Construction Cost Index and the Consumer Price Index.[30,31] Finally, given the importance of choice of interest rate to the magnitude of the amortized costs (and thus unit costs), sensitivity analysis over a range of interest rates (3%−10%) was explored, yielding no appreciable qualitative differences.

MBR and RO processes are relatively expensive compared to the other processes (Table S4). Additionally, economies of scales associated with the larger plants significantly reduce the unit treatment costs for nearly all processes (Table S4). For

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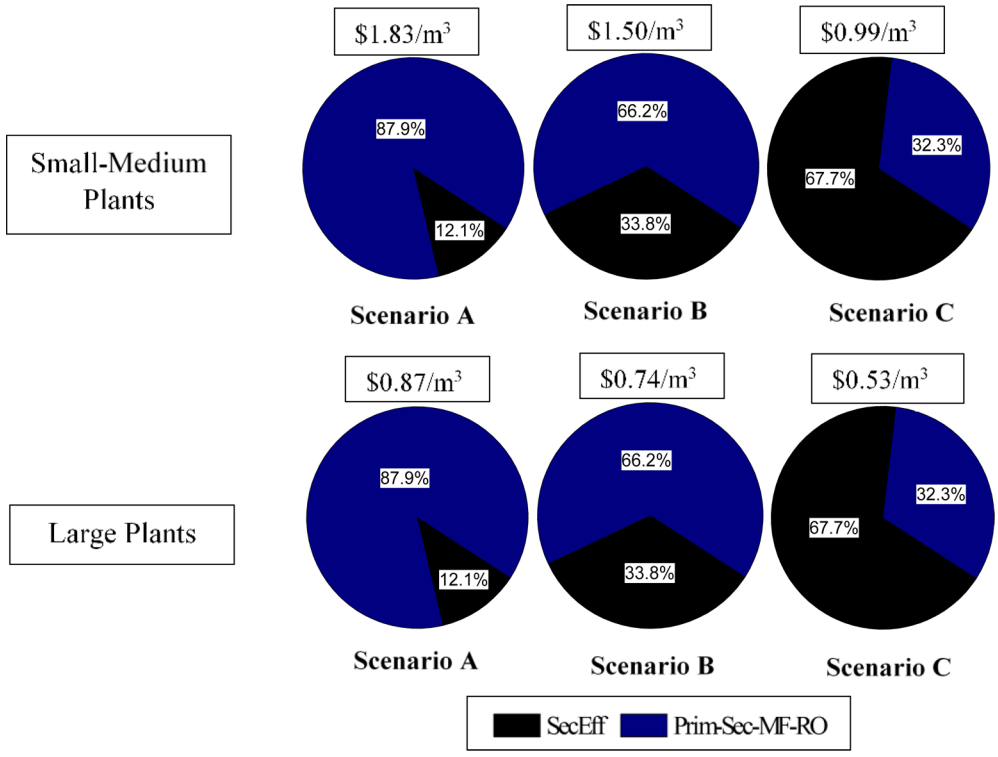


Figure 2. Optimized blending ratios for citrus irrigation from the MF−RO treatment train for small−medium and large treatment facilities; due to TDS restrictions, all model solutions require some degree of desalination (RO). A total of three scenarios were investigated: (A) with crop nutrient and bicarbonate constraints (baseline); (B) without crop nutrient constraints; and (C) without crop nutrient and bicarbonate constraints.

turbid water, a higher dosage of disinfectant is needed to comply with discharge regulations, a step that increases our disinfection costs by 2- to 3-fold.

RESULTS AND DISCUSSION

The RWRM was used to determine the most cost-effective treatment train and associated blending ratios that meet both regulatory standards and crop thresholds. This section presents the costs of each of the treatment trains used to meet irrigation guidelines for citrus and turfgrass. For each crop, three different scenarios were analyzed: (a) with nutrient and bicarbonate constraints (baseline), (b) without nutrient constraints, and (c) without nutrient and bicarbonate constraints. The RWRM screened the treatment systems and eliminated those that did not produce any feasible solution. For example, granular filtration and NF systems were never selected due to their inability to produce blends with sufficiently low ion concentrations (Na^+, HCO_3^-, Cl^-, K^+, and B) to meet the irrigation guidelines for either citrus or turfgrass. A model sensitivity analysis based on plant size, removal performances, and treatment costs is provided in Tables S7−S15.

Treated Wastewater for Citrus Irrigation. Figures 2 and 3 present the results from the RWRM when the treated wastewater from small−medium and large treatment facilities is applied to citrus while meeting Title 22 regulations and citrus irrigation guidelines. Title 22 specifies that recycled wastewater used for orchard irrigation must meet or exceed *undisinfected secondary effluent* water standards.[17,22] Figure 2 presents the blending ratios (in percentage terms) for the least-cost scenarios with the unit costs listed at the top of each pie chart. All solutions contained a certain percentage of desalinated (RO effluent) wastewater due to the excess amounts of P, Na^+, K^+, and CO_3^{-2}. Under baseline conditions

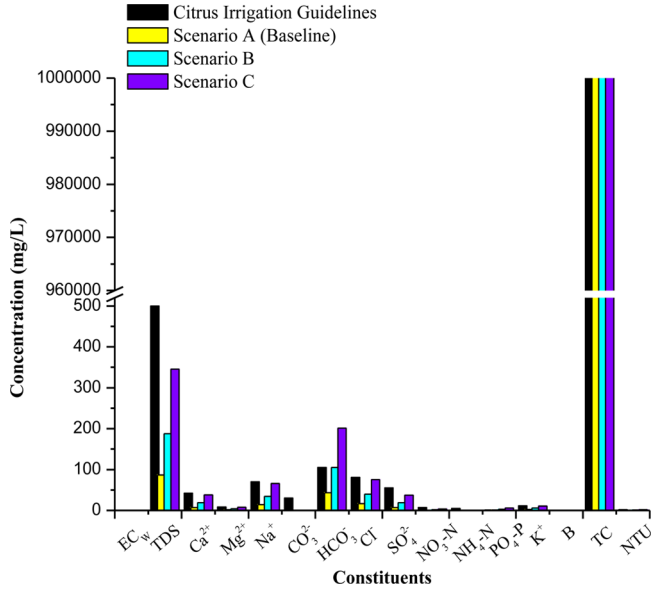


Figure 3. Comparison of irrigation guidelines for citrus (black bars) with the water quality parameters of the different blending ratios from the MF−RO treatment train under the three different constraints: (yellow bars) with crop nutrient constraints, i.e., baseline (RSC = 0.22 mequiv/L); (blue bars) without crop nutrient constraints (RSC = 0.42 mequiv/L); and (purple bars) without nutrient and bicarbonate constraints (RSC = 0.74 mequiv/L).

(scenario A), in which nutrient constraints are applied to maintain NH_4−N, PO_4−P, and K concentrations typically found in irrigation water, the cost-effective solutions were $0.87/m^3 and $1.83/m^3 for large and small−medium plants, respectively; these results were generated from an MF−RO

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Environ. Sci. Technol. 2016, 50, 9390−9399

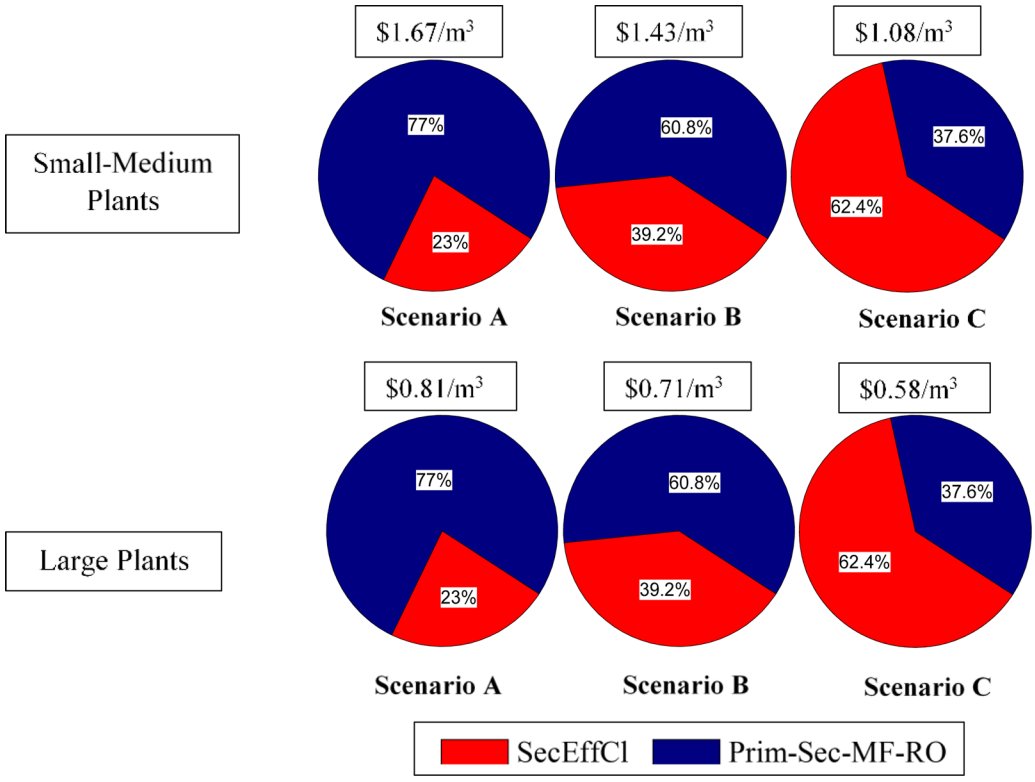


Figure 4. Optimized blending ratios for restricted access turfgrass irrigation with chlorine disinfection from the MF−RO treatment train for small−medium to large treatment facilities; due to TDS restrictions, all model solutions require some degree of desalination (RO). A total of three scenarios were investigated: (A) with crop nutrient and bicarbonate constraints (baseline); (B) without crop nutrient constraints; and (C) without crop nutrient and bicarbonate constraints.

train composed of a blend of 88% RO effluent and 12% secondary effluent.

Figure 3 presents the concentrations of the different water quality parameters for the least-cost MF−RO train (yellow bars) relative to water quality parameters found in typical irrigation water (black bars). As can be seen, the least-cost MF−RO solution produces irrigation water with water quality parameters far superior to crop guideline levels, an outcome driven by the large percentage of RO effluent in the final blended product. For example, TDS concentrations were 83% below the threshold for citrus (86.683 versus 500 mg/L TDS). The most-cost-effective irrigation solution in scenario A is constrained by the amount of phosphorus in the blended product (Figure 3). Although the MF−RO train was the least-cost solution under the baseline (scenario A), $0.87/m^3 and $1.83/m^3 for large and small−medium plants, respectively, are relatively high costs for irrigation water, making this solution challenging in real world applications. For example, agricultural operators in the Westlands Water District, the largest irrigation district in the country, pay around $0.28/m^3 for contract water when it is available[32] but may find water prices on the spot market during drought in excess of $1.62/m^3;[33] alternatively, the price MWD charges its retail agencies for tier 1 water, which is water that comes from the Colorado River Aqueduct or SWP, a cost of $0.76/m^3.[34]

Removing the nutrient concentration limits (scenario B) and bicarbonate constraint (scenario C) leads to a significantly different cost-effective solution. Although the MF−RO train still provides the least-cost solution relative to other trains, removing the nutrient constraint reduces treatment costs by 18%, $0.74/m^3 for large plants and $1.50 for small−medium plants, under scenario B (Figure 2). These lower costs relative to scenario A are directly related to a blend that relies less on RO effluent (25% reduction) and more on secondary effluent. With the removal of the bicarbonate constraint, alternatively, treatment costs under the least-cost MF−RO train are reduced by 46% relative to the baseline ($0.53/m^3 for large plants and $0.99 for small−medium plants), again a result of a blend that relies even less on RO effluent (only 32%) and more on secondary effluent (68%). Importantly, by removing these constraints, the fraction of water treated with the high-cost RO process in the final blended product can be significantly reduced while still meeting water quality guidelines for these crops.

The water quality impacts of relying less on the RO process under scenarios B and C is illustrated by the light blue and purple bars in Figure 3. The concentrations of different constituents increase relative to the baseline scenario (scenario A) yet are still significantly lower than the threshold levels associated with crop requirements. Consequently, even though the overall water quality declines when the fraction of RO effluent decreases, the quality of the blended product still meets citrus irrigation quality requirements at a significantly lower cost. Importantly, in each of the three scenarios, bicarbonate concentrations were below 210 mg/L (Figure 3), well within the range of maximum allowable bicarbonate concentration.[18] The tolerable levels of bicarbonate in irrigation water, expressed as residual sodium carbonate (RSC), are RSC ≤ 1.25 mequiv/L.[9,18] In all cases, the blended effluents were safe for citrus and soils, with RSC values of <0.75 mequiv/L for both citrus and turfgrass.

DOI: 10.1021/acs.est.6b02073
Environ. Sci. Technol. 2016, 50, 9390−9399

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There are two final observations worth noting. First, the treatment costs for large facilities were nearly 50% less than the costs under small−medium plant sizes, highlighting the role that economies of scale can play in making treated wastewater more competitive for implementation in real-world applications. Second, the results from blending the effluent from different treatment processes generated a feasible and cost-effective irrigation source for citrus while complying with Title 22 regulations and without affecting soil and crop health. However, significant numbers of total coliform (TC) bacteria (10^6 MPN/100 mL) are projected to be present in the effluents (Figure 3). Therefore, appropriate precautions should be taken when applying this water to citrus grooves.

Treated Wastewater for Restricted Turfgrass Irrigation. Figures 4 and 5 present the results from RWRM when the treated wastewater from small−medium to large treatment plants is used for turfgrass irrigation. Under Title 22 regulations, restricted access turfgrass requires at least disinfected secondary-treated water.[17,22] Although many turfgrass fields (such as golf courses) do not typically use desalinated wastewater for irrigation, if the treated wastewater contains elevated salinity levels, these fields require periodic flushing with fresh water to remove accumulated salt.[9] Under the baseline (scenario A), the MF−RO−Cl train with a blend of 77% MF−RO effluent and 23% secondary effluent with chlorine disinfection was the lowest-cost solution at \$0.81/m^3 and \$1.67/m^3 for large and small−medium plants, respectively (Figure 4). Notable is the superior quality of the blended water relative to the parameters under the turfgrass irrigation guidelines, specifically in TDS and salts, as shown in Figure 5 (black bars), a consequence of the significant RO fraction in the final blend.

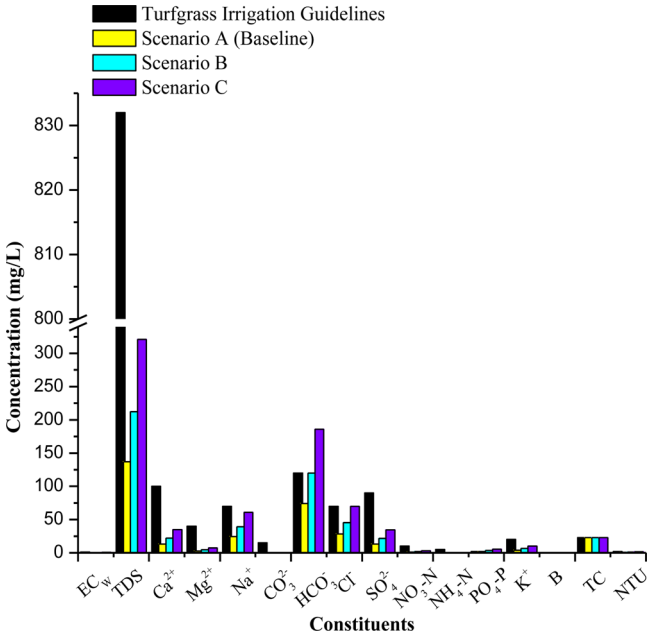


Figure 5. Comparison of irrigation guidelines for restricted access turfgrass (black bars) with the water quality parameters of the different blending ratios from the MF−RO treatment train under the three different constraints: (yellow bars) with crop nutrient constraints, i.e., baseline (RSC = 0.32 mequiv/L); (blue bars) without crop nutrient constraints (RSC = 0.47 mequiv/L); and (purple bars) without nutrient and bicarbonate constraints (RSC = 0.69 mequiv/L).

When the nutrient constraint was removed, the low-cost treatment train is again MF−RO−Cl, with a blend consisting of 61% MF−RO effluent and 39% secondary effluent with chlorine disinfection. Compared to the baseline, RO effluent was reduced by approximately 21% while the amount of secondary effluent was almost doubled, resulting in a total treatment cost of \$0.71/m^3 and \$1.43/m^3 for large and small−medium plants, respectively (Figure 4). Even with the large reduction in the percentage of RO effluent, which was replaced with secondary effluent, the resulting irrigation water quality significantly exceeds irrigation guidelines for turfgrass (Figure 5). When the bicarbonate constraint was removed, the RO portion of the blended water was further reduced to 38% (a 38% decrease from scenario B), and the portion of secondary effluent increased to 62% (a 60% increase from scenario B), with a final treatment cost of \$0.58/m^3 for large plants and \$1.08/m^3 for small−medium plants.

In each of the solutions under this scenario, chlorine disinfection was necessary to ensure that the water quality was compliant with Title 22 regulations. Also, and similar to the citrus case above, economies of scale reduced the treatment costs associated of larger plants by up to 50% relative to those of the small−medium sized plants.

Treated Wastewater for Unrestricted Turfgrass Irrigation. The major difference under Title 22 between recycled wastewater used to irrigate turfgrass with restricted access and that with unrestricted access is that wastewater used to irrigate the latter must be irrigated with at least disinfected tertiary-treated wastewater with a total coliform count below 2.2 MPN/100 mL.[17,22] The RWRM found that the MF−RO−Cl train produced the least-cost solution, with a cost of \$0.86/m^3, \$0.80/m^3, and \$0.72/m^3 for scenario A, B, and C, respectively, for large plant sizes. Treatment costs under the small−medium plants sizes were nearly double those relative to the larger plants (\$1.76/m^3, \$1.56/m^3, and \$1.27/m^3 for scenarios A, B, and C, respectively). A detailed analysis of treated wastewater for unrestricted turfgrass irrigation can be found in the Supporting Information.

Treated Wastewater versus Traditional Irrigation Sources. The blending method used in this work produced alternative irrigation options in place of the traditional irrigation sources, which typically rely solely on secondary and tertiary effluent and natural sources.[5] The water quality of surface and groundwater sources is typically better than conventionally treated wastewater, with low electrical conductivity (typically less than 0.522 dS/cm) and low ion concentrations (0.1 mg/L B, 23.6 mg/L Na^+, 28 mg/L Cl^-, and 88.3 mg/L Ca^{2+}).[35] Secondary and tertiary effluents have relatively high salinity, with TDS in excess of 500 mg/L, electrical conductivity exceeding 0.85 dS/m, and high ion concentrations (0.3 mg/L B, 95.9 mg/L Na^+, 130 mg/L Cl^-, and Ca^{2+} in excess of 49 mg/L)[20,21] (Table S2). Compared to natural water sources, the blended effluent provides competitive or even superior water quality at an affordable cost. While groundwater pumping costs are relatively cheap (\$0.115/m^3), groundwater supplies are often under stress or may be entirely unavailable.[36−39] Furthermore, as groundwater levels drop, the cost of bringing this water to the surface increases, incentivizing irrigators and municipalities to look elsewhere for their water supplies.[38,40]

While disinfected secondary and tertiary effluents have been used to irrigate landscapes in restricted and unrestricted areas for some time, particularly on golf courses, the sustainability of such practices is sometimes problematic. That is, the salinity of

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Environ. Sci. Technol. 2016, 50, 9390−9399

certain conventionally treated wastewater is above the maximum irrigation tolerance for turfgrass, which, over time, harms both the soil and the crop due to salt accumulation.[9,28,41] The results from our analysis suggest the possibility of a high-quality irrigation source at a competitive price relative to what is typically observed today. On the basis of the literature, we determined that the cost of disinfected secondary and tertiary (granular filtration) effluents is around $0.25/m^3 and $0.19/m^3 (thus, the total cost of tertiary effluent is $0.25 + $0.19 = $0.44/m^3), respectively (Table S4), whereas the cost-effective blending ratio for turfgrass from our model was $0.58/m^3 (restricted access; large plants) and $0.72/m^3 (unrestricted access; large plants). Our low-cost, higher-quality solution is achieved through the use of secondary effluents in the blend, with chlorine disinfection of the final product ensuring compliance with Title 22 regulations.

Nutrients in Treated Wastewater and Cost Savings Associated with Lower Synthetic Fertilizer Usage. In addition to providing optimal irrigation solutions to citrus and turfgrass, reusing wastewater for agriculture also offers cost savings on synthetic fertilizers due to the appreciable amount of nutrients present in the wastewater streams (PO_4−P and K). To illustrate this, we assume synthetic fertilizers costs of $0.86/kg nitrogen (N), $1.19/kg phosphate ($P_2O_5$), and $0.72/kg potash ($K_2O$), based on average wholesale prices[45−47] and irrigation rates of 2.54[48] and 5.08 cm/week for citrus and turfgrass, respectively.[49,50] For citrus, recommended fertilizer application rates to meet plant requirements using synthetic fertilizer alone range from 100 to 400, 0−208, and 135−224 kg/ha-year for N, P, and K, respectively, resulting in fertilizer costs of $777.94/ha-year.[42−44,51−53] For turfgrass, the recommended application rates range from 98 to 195, 49, 146 kg/ha-year for N, P, and K, respectively, for a cost of $332.08/ha-year.[42−44,54−57] Before presenting the cost comparisons, though, it should be noted that phosphorus demand by citrus is not uniform and largely depends on the available phosphorus in the soil, evaluated by measuring phosphorus in both soil and leaves.[42,43] Thus, the recommended phosphorus fertilization rate can range from 0 to 228 kg P_2O_5/ha-year, depending on the age of the trees and assuming fruit production of 1483 boxes/ha-year.[44]

Now we consider the degree to which the nutrient loads from our cost-effective solutions may substitute for synthetic fertilizers. Tables S5 presents the list of nutrient loads corresponding to each cost-effective blend. For the three different blending solutions identified by the model (corresponding to scenarios A, B, and C), treated wastewater under scenario C provides the largest concentration of nutrients to citrus (2.00, 172, and 174 kg/ha-year of N, P, and K, respectively (Table S6)). Comparing these rates with the recommended rates above, we see that phosphorus concentrations in the blended irrigation water may meet, and in some cases exceed, the crop's demand for phosphorus. Because NH_4−N removal during the wastewater treatment process is so effective, little NH_4 is available as fertilizer. Given these contributions, fertilizer costs can be reduced by $33.134/ha-year under scenario C. Under scenario B, treated wastewater can provide 1.32, 87.2, and 90.8 kg/ha-year of N, P, and K, respectively, providing cost savings of $170.41/ha-year, while under scenario A, the cost savings can be as high as $67.15/ha-year (Table S6). It must be stated here that some wastewater treatment plants now include a biological phosphorus removal step, which removes nearly all phosphorus from the waste stream. Thus, irrigation water generated from these plants will not have significant P concentrations.

For turfgrass, the irrigation water generated under scenarios B and C had phosphorus and potassium concentrations that exceeded average demand for turfgrass (Table S5). Under scenario A, the annual value stemming from the presence of nutrients in the irrigation water was $95/ha-year, replacing 71% of the cost of synthetic fertilizer (Table S6); synthetic fertilizer requirements were based on an average value across different turfgrass species with different N, P, and K tolerances and requirements.[54,55] Consequently, the concentrations of potassium and phosphorus in our solutions may be too high for low-tolerance strains. For instance, for sand-based soil Kentucky blue grass and perennial ryegrass (cool-season lawn), the ideal potash application ranges from 195 to 390 kg/ha, whereas bentgrass and fine fescue lawns only require 98 kg/ha of potash annually.[56,57] This problem can be solved by reducing potassium and phosphorus concentrations in the final blended product through increasing the RO portion of the irrigation water, a solution that will increase the cost of the irrigation water.

The RWRM was shown to optimize blending combinations of different treated municipal wastewater effluents to provide irrigation water tailored to meet crop requirements at minimum costs. Alternative scenarios related to water quality parameter thresholds and guidelines were evaluated to demonstrate the model's flexibility and usefulness; in all cases, a certain fraction of the water required desalination to reduce salinity in the final blended product. When more restrictions were placed on irrigation water quality, the model responded with a large portion of RO water in the final blended product, which resulted in a significantly higher treatment cost. With the gradual relaxation of each of the constraints, the RO portion of the final blended product was reduced, lowering the treatment costs to a level appropriate for adoption.

Limitations. While there were a number of simplifying assumptions in the model presented above, two in particular are worth discussing. First, we assumed a cost-minimization framework that identified the least-cost solution to produce treated wastewater given particular water quality constraints. This framework overlooks possible solutions in which the users of the water may be willing to accept poorer quality water in certain dimensions (along with the consequent yield reductions) for a lower price. To represent this alternative, crop-water production functions could be incorporated into the model along with their associated prices and costs so that the model could identify efficient solutions in which net benefits are maximized; net benefits are defined by the profits to the water user less the costs to the wastewater treatment plant operator. Crop-water-salinity production functions developed and utilized in Letey et al. (1985), Kan et al. (2002), and Schwabe et al. (2006) are being further developed for such an extension.[58−60] Second, our model focuses on the water quality requirements associated with two separate crops in isolation. In reality, demand for the treated municipal wastewater nor the current distribution will likely allow wastewater treatment managers to produce water for a single type of crop or product. There are a number of ways the model could be adjusted to reflect this, including incorporating the net benefits framework mentioned above. Alternatively, the current cost-minimization framework could be modified to generate solutions subject to constraints imposed on the most-sensitive parameters across the array of crops. Because this model is intended to be flexible

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DOI: 10.1021/acs.est.6b02073
Environ. Sci. Technol. 2016, 50, 9390−9399

and allow for the evaluation of a wide variety of treatment processes and output scenarios, such explorations are easily incorporated into the current framework.

Using the RWRM, wastewater treatment trains can be optimized to produce irrigation water suitable for a wide range of crops with varying salinity tolerance, reducing the impact on soil and crop quality that is currently experienced by irrigators using conventionally treated wastewater. Salinity, heavy metals, and pathogens were minimized to comply with existing regulations and safe agriculture practices. With the utilization of this blending technique as an alternative irrigation source for agriculture, freshwater resources would be reserved to cope with drought-induced extreme water scarcity.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.6b02073.

Tables showing the average removal rates (%) of contaminants by each treatment process relative to plant influent, a summary of treated municipal wastewater quality after each treatment process, guidelines of water quality for irrigation water of specific crops, average individual treatment costs of different wastewater processes, fertilizers supplied using blended irrigation sources, costs and cost savings on synthetic fertilizer using blended irrigation sources, sensitivity analysis on removal performances and treatment cost for citrus, sensitivity analysis on cost of disinfection of citrus irrigation water, sensitivity analysis on treatment cost and removal performances for turfgrass with restricted access, sensitivity analysis on cost of disinfection of turfgrass irrigation water, sensitivity analysis on treatment cost and removal performances for turfgrass with unrestricted access, sensitivity analysis on cost of disinfection of turfgrass irrigation water, parameters and cost of a full-scale NF process, parameters and cost of full-scale MF and UF processes, treatment cost of a full-scale ozonation process, treatment cost of a full-scale chlorine disinfection process, treatment costs of different wastewater treatment processes, average individual treatment costs of different wastewater processes with time adjustment. Figures showing optimized blending ratios for unrestricted access turfgrass irrigation and a comparison of irrigation guidelines for unrestricted access turfgrass (black bars) with the water quality parameters of the different blending ratios. (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research is funded by the USD/AFRI Grant Program (2015-68007-23191) and the NIFA-BARD Program (project no. NB-8307-14).

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